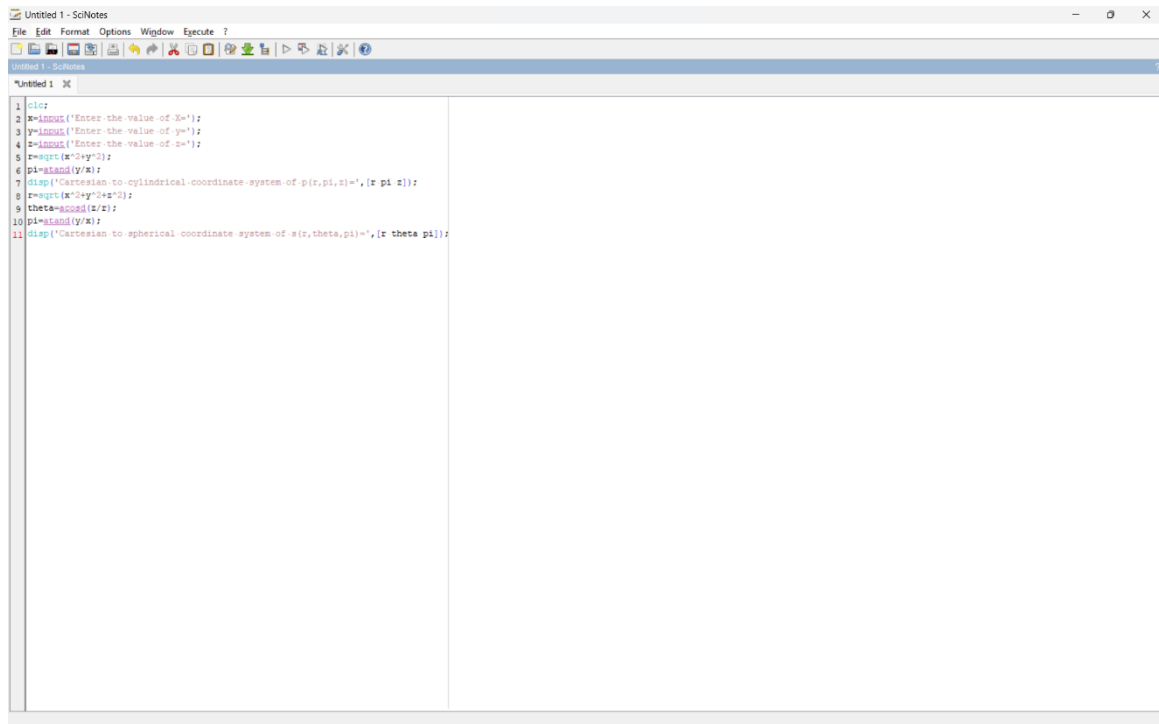
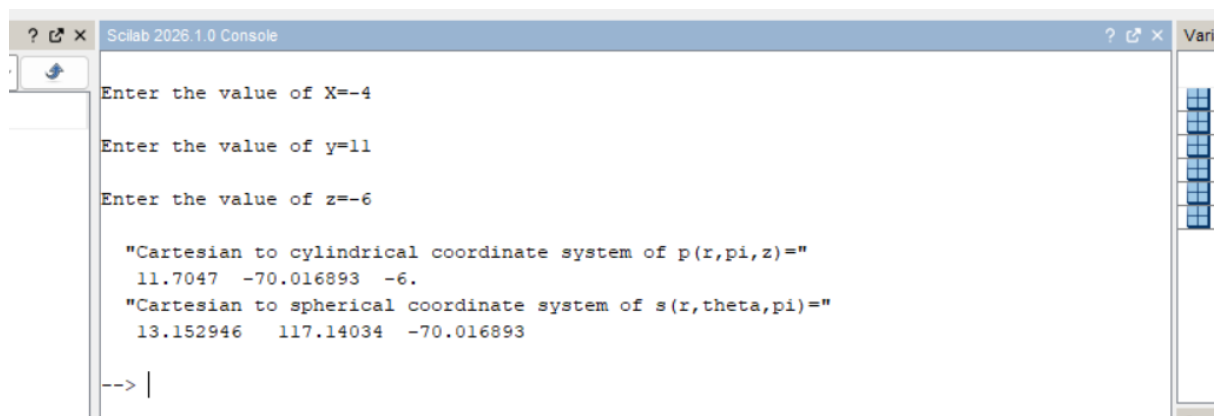


EXP 1: Transformation between Cartesian, Cylindrical and spherical coordinate system

Test case (i) Transformation of Cartesian coordinate points in to Cylindrical and spherical coordinate system



```
1 clear
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atan2(y/x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]');
8 r=sqrt(x^2+y^2);
9 theta=atan2(z/z);
10 pi=atan2(y/x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]');
```



```
SciLab 2026.1.0 Console
Enter the value of X=-4
Enter the value of y=11
Enter the value of z=-6

"Cartesian to cylindrical coordinate system of p(r,pi,z)="
11.7047 -70.016893 -6.
"Cartesian to spherical coordinate system of s(r,theta,pi)="
13.152946 117.14034 -70.016893
--> |
```

EMF OBSERVATION

x, y, z

$$x = -4$$

$$\rho = \sqrt{x^2 + y^2}$$

$$y = 11$$

$$\phi = \tan^{-1}(y/x)$$

$$z = -6$$

$$z = z$$

cylindrical

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} \Rightarrow \sqrt{(-4)^2 + (11)^2} \\ &= \sqrt{16 + 121} \\ &= \sqrt{137} = 11.7 \end{aligned}$$

$$\tan^{-1}\left(\frac{11}{-4}\right) = \cancel{10}^\circ \rightarrow 109.9^\circ$$

$$z = z \Rightarrow z = -6$$

cylindrical

$$(11.7, \cancel{10}^\circ, -6)$$

$$\begin{aligned} \gamma &= \sqrt{x^2 + y^2 + z^2} = \sqrt{(-4)^2 + (11.7)^2 + (-6)^2} \\ &= \sqrt{16 + 121 + 36} \\ &= 13.15 \end{aligned}$$

spherical

$$\begin{aligned} \theta &= \cos^{-1}\left(\frac{z}{\gamma}\right) = \cos^{-1}\left(\frac{-6}{13.15}\right) \\ &= \cancel{10}^\circ 117.16^\circ \end{aligned}$$

spherical

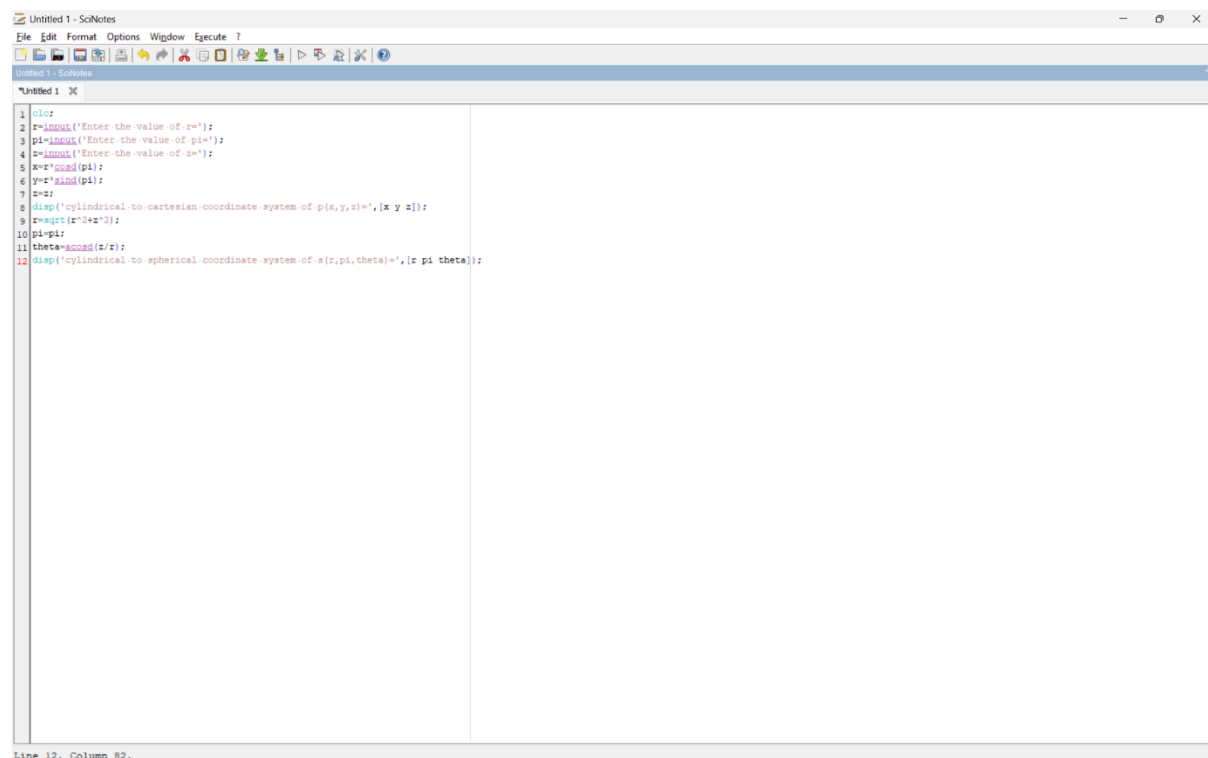
$$(\gamma, \theta, \phi)$$

$$(13.15, \cancel{10}^\circ, \cancel{10}^\circ)$$

$$117.16 \quad 109.9 \rightarrow 70^\circ$$

(i)

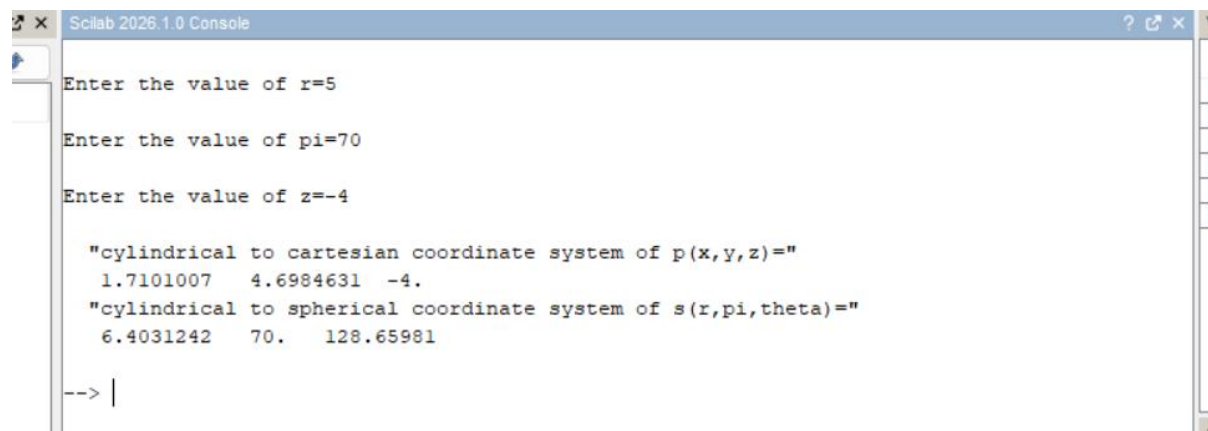
Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system



The image shows the Scilab 2026.1.0 Editor window with a script titled 'Untitled 1 - Scilab'. The script defines a function 'cylindrical_to_cartesian' and 'cylindrical_to_spherical'. It prompts the user to enter values for r, pi, and z, then calculates the Cartesian coordinates (x, y, z) and the spherical coordinates (s, pi, theta).

```
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 z=input('Enter the value of z=');
5 x=r*cos(pi);
6 y=r*sin(pi);
7 z=z;
8 disp('cylindrical to cartesian coordinate system of p(x,y,z)=[x y z]');
9 z=sqrt(r^2+z^2);
10 pi=pi;
11 theta=acos(z/r);
12 disp('cylindrical to spherical coordinate system of s(r,pi,theta)=[r pi theta]');
```

Line 12, Column 82.



The image shows the Scilab 2026.1.0 Console window displaying the output of the script. It shows the prompts for r, pi, and z, followed by the calculated Cartesian and spherical coordinates.

```
Enter the value of r=5
Enter the value of pi=70
Enter the value of z=-4

"cylindrical to cartesian coordinate system of p(x,y,z)="
1.7101007  4.6984631  -4.
"cylindrical to spherical coordinate system of s(r,pi,theta)="
6.4031242  70.  128.65981

--> |
```

convert

$$\rho = 5$$

$$\phi = 70^\circ$$

$$z = -4$$

cylindrical $\rightarrow$ Rectangular

$$x = \rho \cos \phi = 5 \cos 70^\circ = 5(0.34202) \\ = 1.7101$$

$$y = \rho \sin \phi = 5 \sin 70^\circ = 5(0.939693) \\ = 4.6985$$

$$z = -4$$

$$(x, y, z) = (1.7101, 4.6985, -4)$$

cylindrical $\rightarrow$ spherical

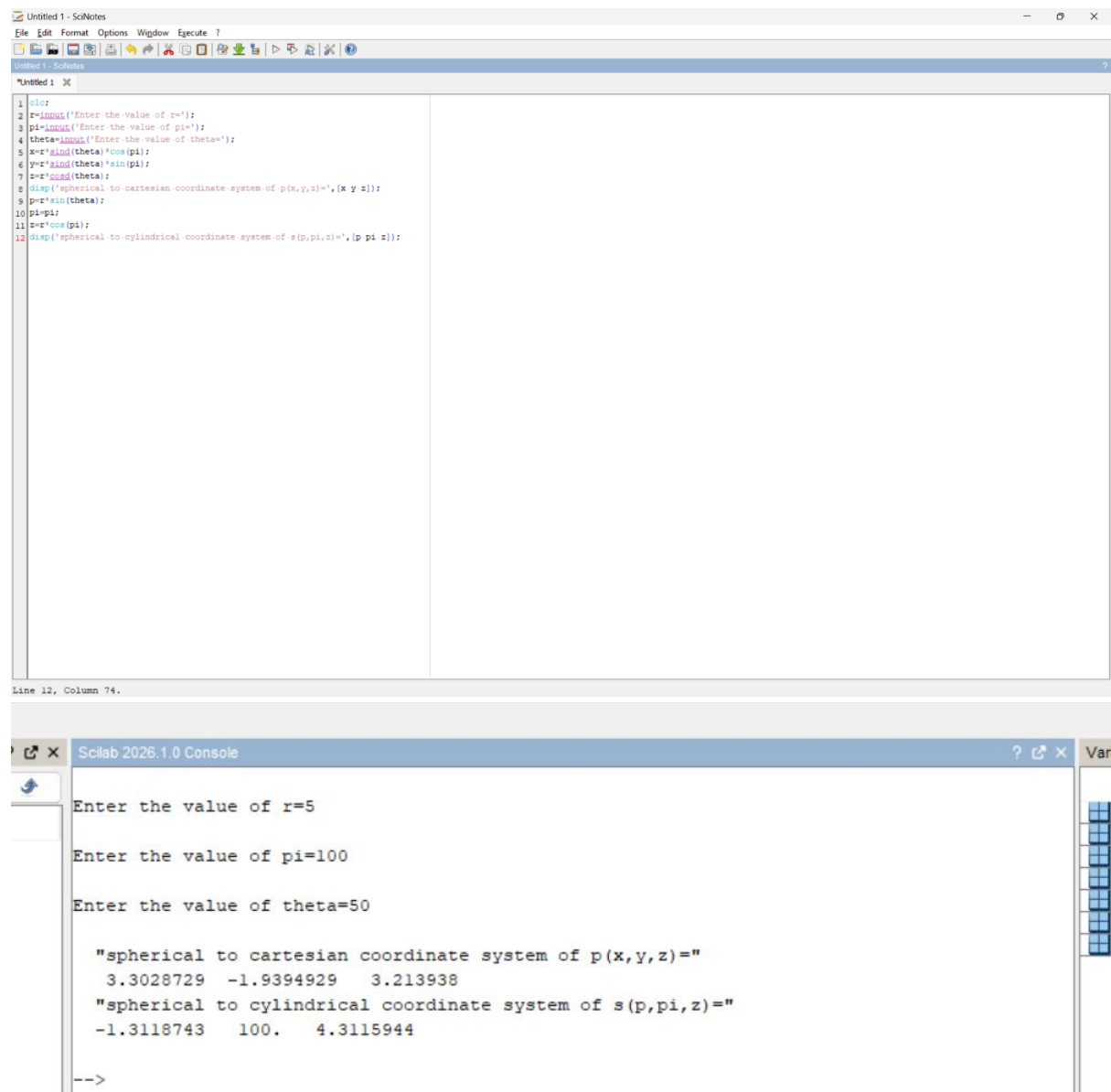
$$R = \sqrt{x^2 + y^2 + z^2} \Rightarrow \sqrt{1.7101^2 + 4.6985^2 + (-4)^2} = \sqrt{25} = 5$$

$$\theta = \cos^{-1} \left(\frac{z}{R} \right) \Rightarrow \cos^{-1} \left(\frac{-4}{5} \right) = \cos^{-1} (-0.8) \\ = 128.66^\circ$$

$$\phi = 70^\circ$$

$$(R, \phi, \theta) = (5, 70^\circ, 128.66^\circ)$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system



The screenshot displays the Scilab 2026.1.0 environment. The main window shows a script titled 'Untitled 1 - Scilab' with the following code:

```
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 theta=input('Enter the value of theta=');
5 x=r*cos(theta)*cos(pi);
6 y=r*cos(theta)*sin(pi);
7 z=r*sin(theta);
8 disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z]');
9 p=[x y z];
10 pi=pi;
11 z=r*cos(pi);
12 disp('spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z]');
```

The console window, titled 'Scilab 2026.1.0 Console', shows the execution of the script with the following input and output:

```
Enter the value of r=5
Enter the value of pi=100
Enter the value of theta=50

"spherical to cartesian coordinate system of p(x,y,z)="
 3.3028729 -1.9394929  3.213938
"spherical to cylindrical coordinate system of s(p,pi,z)="
-1.3118743  100.  4.3115944

-->
```

code (ii)/

Rectangle

$$r = 5$$

$$\theta = 50^\circ$$

$$\phi = 100^\circ$$

$$x = r \sin \theta \cos \phi = 5 (\sin 50^\circ) (\cos 100^\circ) \\ = 5 (0.7660) (-0.1736) = -0.665$$

$$y = r \sin \theta \sin \phi = 5 (\sin 50^\circ) (\sin 100^\circ) \\ = 5 (0.7660) (0.9848) = 3.770$$

$$z = r \cos \theta = 5 (\cos 50^\circ) = 3.213$$

$$(x, y, z) = (-0.665, 3.770, 3.213) \rightarrow \text{degree} \\ = (-1.93, 3.77, 3.21) \rightarrow \text{rad}$$

cylindrical

$$\rho = r \sin \theta = 5 \sin (50^\circ)$$

$$= 5 (0.7660)$$

$$= 3.8302$$

$$\phi = 100^\circ$$

$$z = r \cos \theta = 5 \cos 50^\circ \\ = 3.21$$

$$(r, \phi, z) = (3.83, 100, 3.21) \rightarrow \text{degree} \\ (-1.3, 100, 4.311) \rightarrow \text{rad}$$